Preference-Graph Repair and Retrieval Utility: An Empirical Opportunity Audit

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Abstract

Preference-graph repair optimizes structural consistency, while retrieval evaluation measures relevance at ranking positions. We investigate where these responses separate in preserved retrieval evidence, holding candidates, comparisons, extraction and evaluation fixed across unrepaired, greedy and exact deletion branches. The empirical contribution is a decomposition of observed intervention, ranking and measured-relevance responses, with their dataset and provider-availability composition. In a selected cache of 213 queries, greedy and exact repair change 180 and 160 rankings, respectively, but change measured utility in only 37 and 25 cases. Among changed rankings in positive-containing pools, 74 cases per branch preserve the evaluated top-ten grade sequence; the branches share queries. Such cases occur in all four datasets and both availability strata, whereas zero-positive pools are concentrated in two datasets. The distinction separates missing judged opportunity from active ranking changes that the metric cannot distinguish. Corrected query-block inference retains zero Holm rejections in 35 exact-versus-greedy comparisons, but only 4–99 query sums per endpoint are nonzero. The completed compact-model cohort has no strict cyclic-positive case and substantial instrument limitations. A complete six-graph diagnostic additionally finds utility variation among equal-cost repairs in one graph. These conditional distributions and finite diagnostics extend earlier construction-dependent findings without introducing a new repair algorithm. Selection, incomplete judgments and inferential assumptions limit generalization; nonsignificance does not establish equivalence.

Keywords: preference graphs, feedback arc set, graph repair, rank aggregation, information retrieval, retrieval evaluation

1 Introduction

Pairwise comparisons provide a natural interface for ranking documents: an observation that document u should precede v becomes a directed edge $u \to v$. Comparisons from several rankers or language models can disagree and form cycles. Removing a feedback arc set restores acyclicity, so the retained preferences admit a consistent order. This structural success is observable even when no relevant document receives a better evaluated position. The distinction matters for data science pipelines that optimize an internal graph objective but are deployed against external, task-dependent outcomes. In retrieval, repairing disagreement with the input preferences must therefore be evaluated alongside relevance at the output ranking.

A retrieval pipeline contains several opportunities for an intervention to matter. Candidate selection determines which documents are available; comparison collection supplies evidence; graph construction retains and weights it; repair modifies the graph; extraction produces an ordering; and relevance judgments determine the measured effectiveness. A failure at one stage can make a later comparison uninformative. Conversely, finding cycles and deleting edges establishes that an intervention acts, but not that it improves relevance. These possibilities require separate observations rather than an inference from one aggregate metric.

1.1 Research question and scope

We ask: at which stages do structural changes fail to affect measured retrieval utility in the evaluated preference-graph pipelines, and what do saved rankings and relevance grades explain about those outcomes? Our primary contrast holds candidate documents, preference evidence, extraction, tie handling and evaluation fixed while changing only the repair branch. It therefore differs from comparing complete ranking systems with different acquisition or aggregation procedures.

The evidence combines four score-derived benchmarks, selected cached multi-provider judgments, an independent compact-model collection and completed external controls. Repeated construction regimes and model observations do not increase the independent query sample. The larger selected cache provides informative cyclic graphs, while the prospective strict construction has almost no cyclic graph containing a judged-positive candidate. Neither can substitute for the other's sampling and measurement contract.

The missing empirical knowledge is how frequently the intervention stops at each stage, and whether superficially similar zeros have the same composition. A cyclic graph with an unchanged extracted order, a changed order with no judged-positive candidate, and a changed positive-pool order with fixed evaluated gain positions answer different evaluation questions. An average effect or a count of nonsignificant tests cannot distinguish them. We use complete saved permutations and relevance maps to make those distinctions observable, then examine their composition rather than presenting a metric identity as a discovery.

Exact optimization controls deletion cost; a complete small-instance diagnostic tests utility ambiguity among equal-cost repairs. Both support the conditional audit. All prospective results in this article refer to the already completed 20-query collection; this revision collects no new judgments and uses no outcomes from a separate replication.

1.2 Contributions

The contribution is additional empirical knowledge about the transmission of an intervention to an evaluated outcome:

1. **Locate the inactive stage.** Cohort-specific structural and ranking counts distinguish construction inactivity from an extractor's unchanged response (Table 2). The canonical and selected-cache populations support different decompositions and are not interchangeable.
2. **Explain and disaggregate measured zeros.** Saved-ranking attribution separates zero-positive pools from fixed positive-pool grade profiles, while retaining gains and harms (Table 3). Dataset and availability-stratum counts show where each mechanism occurs and expose small or empty cells (Table 4; Online Resource 1). This composition, rather than the invariance identity alone, is the central extension of earlier evidence.
3. **Bound what structural exactness identifies.** All-optimum utility ranges for six authentic graphs show that one returned structural optimum need not determine utility, while establishing neither prevalence nor a deployable selector (Table 8).

The evaluation consequence is to report the opportunity denominator alongside an effect: a null from an inactive or unreliable instrument cannot establish efficacy among informative cyclic-positive cases, and an unchanged measured grade profile cannot establish semantic irrelevance. The reporting prescription is not a validated predictor of benefit.

Graph-repair algorithms and ranking–MWFAS theory are established prior work. The broad construction-dependent retrieval message is already in the public preprint [30]; benchmark measurements and previously released resources are reused. Corrected inference repairs

the independent-unit analysis (Section 4.3), and reconstruction verifies the processed evidence (Section 4.4); neither alone is a novelty claim. The new attribution and composition are analyses of preserved observations, not new independent datasets or experiments.

2 Related Work

2.1 Preference aggregation and graph repair

Probabilistic aggregation and combinatorial repair answer different questions about pairwise evidence. Bradley–Terry models estimate latent strengths from comparisons [5, 17]. Rank Centrality estimates strengths using a comparison Markov chain [24]. Combinatorial Hodge theory separates ranking and cyclic components of pairwise flow [19]. These approaches motivate alternative extraction controls; their assumptions do not make arbitrary edge-deletion costs calibrated correctness probabilities.

Rank aggregation and minimum-disagreement ordering have a substantial combinatorial literature [2, 12]. Feedback arc set optimization removes edges so that the retained directed graph is acyclic; exact methods and linear-ordering formulations are established [3, 14]. We use certified optimization as a diagnostic control on a fixed objective, not as a new formulation or solver.

The published training-free acyclic-ranking pipeline already develops the ranking–feedback-arc relationship and combines pruning, cycle-safe add-back and refinement [31]. Those components and their ranking-recovery results are prior work. The present comparison instead measures the marginal effect of deletion against external retrieval judgments. A separate related graph-benchmark manuscript concerns graph identity, structural method comparison and reproducibility; its benchmark and numerical results are not contributed here. Its status and relationship are disclosed confidentially to the editor.

SpringRank infers real-valued hierarchy through a weighted quadratic energy and a linear system, without requiring edge deletion [11]. It is a useful direct-aggregation control on the same original graph. Comparing repair plus Copeland with SpringRank changes the extractor, however, and cannot identify the isolated effect of repair. That distinction also applies to graph-free fusion: reciprocal rank fusion combines ranks without constructing the preference graph [10].

2.2 Retrieval utility and related empirical audits

Pairwise ranking prompting (PRP) studies language-model comparisons as a document-ranking interface [25]. PRP-Graph additionally uses target-label probabilities in graph construction and aggregation [21]. Our adapted author-code comparison supplies a relevant retrieval baseline, while retaining the distinction between its acquisition/aggregation contract and fixed-judgment repair. A win-only historical cache cannot recover the label probabilities required by that method. Listwise ranking systems elicit an ordering jointly rather than one pair at a time [27]. Setwise prompting selects among several documents and compares pointwise, pairwise and listwise elicitation within a common evaluation framework [36]. Pointwise prompts score documents individually; setwise selection and listwise ordering change the elicited evidence and aggregation contract. These distinctions complement our fixed-evidence contrast rather than constitute additional tested baselines.

LLM-RankFusion mitigates presentation-order inconsistency through prompting and calibration, and addresses transitive inconsistency through aggregation [35]. Preference Graph Ensemble and Denoise combines multiple evaluators to construct acyclic judgments for model evaluation, response selection and training-data selection [16]. TourRank uses tournament-style grouping and a points-based ensemble for document ranking [8]. These methods demonstrate that consistency mitigation can be useful under their acquisition and aggregation contracts. They do not establish that minimum-weight edge deletion improves external relevance when those contracts and the extractor are held fixed.

Xu et al. demonstrate nontransitivity-induced sensitivity in model evaluation and investigate tournament acquisition with Bradley–Terry aggregation [33]. Their endpoint is model-ranking reliability. Here the units are document-retrieval queries, and external relevance judgments evaluate edge deletion with acquisition and extraction

fixed. Neither result transfers automatically to the other task.

The distinction between intrinsic consistency and task-dependent quality is familiar in data-quality research [32]. Acyclicity and minimum disagreement describe the preference evidence; normalized discounted cumulative gain evaluates graded relevance at ranking positions [18]. The present contribution is not the mathematical possibility of disagreement between these objectives. It is the measured frequency and attribution of inactivity across controlled retrieval pipelines, together with exact-cost and optimum-choice diagnostics.

Incomplete assessments are a separate established measurement limitation. Buckley and Voorhees study retrieval evaluation when relevance information is incomplete [6]. That work grounds our distinction between an evaluation zero and semantic irrelevance; it does not validate these selected cohorts or imply that another metric would recover their unobserved relevance. We retain the original nDCG contracts rather than introducing a post-hoc metric replacement.

The public retrieval preprint [30] already separates construction, repair and extraction and reports conditional retrieval effects. We do not claim that broad message anew. The present increment is the quantified location and composition of opportunity losses, complete selected-cache relevance-profile attribution and the finite all-optimum diagnostic, supported by corrected query-block inference. Online Resource 1 maps the preprint's sections/tables to reused evidence, changed construction/normalization, added analyses and superseded interpretations. Its legacy cyclicity summaries and the current retention-matched protocol are different pipeline versions; an execution-level bridge for every old row is not established. Reused evidence is not independent replication. Neither published algorithms [31] nor related structural benchmarks' methods/results are new contributions here. Multiplicity-aware paired inference and the distinction between non-significance and equivalence remain essential [7, 20, 29].

3 Problem Formulation

3.1 Fixed-evidence repair contrast

For query q, let D_q be a fixed candidate set and $G_q = (D_q, E_q, w_q)$ its directed preference graph. Edge $u \to v$ favors placing u ahead of v; its nonnegative weight represents recorded evidence strength, not ground-truth relevance. A feedback arc set F makes $G_q - F$ acyclic. Minimum-weight repair solves

$$F_q^\star \in \arg\min_{F \subseteq E_q} \sum_{e \in F} w_q(e) \quad \text{subject to } G_q - F \text{ being acyclic.} \tag{1}$$

The empty deletion is optimal for an acyclic positive-cost graph. Acyclicity need not imply a unique total order or relevance-optimal preferences. A structural optimum need not identify a unique deletion set either.

Let $\mathcal{R}$ denote a fixed deterministic ranking extractor and U_q the retrieval metric with fixed relevance judgments and normalization. For a returned repair F, the observed contrast is

$$\Delta_q(F) = U_q(\mathcal{R}(G_q - F)) - U_q(\mathcal{R}(G_q)) \,. \tag{2}$$

Greedy, exact and unrepaired branches share these inputs. An exact certificate concerns Eq. (1), not the maximum of Eq. (2). It also does not characterize every equally optimal repair or all rankings obtainable from a repaired graph.

3.2 Rankings, relevance and opportunity stages

The principal extractor in the cached-judgment audit is unweighted Copeland: outgoing degree minus incoming degree, sorted in descending order with ascending document-ID ties. The score-derived studies also hold other extractors fixed within their repair contrasts. Some extractors can be applied to cyclic graphs; topological extraction requires a directed acyclic graph (DAG) and appears only in suitable repaired-graph comparisons. Acyclicity alone does not guarantee that a score-based extracted order respects every retained edge.

Let $\pi_q = (d_1, \ldots, d_{P_q})$ rank the fixed candidates, and let $r_q(d)$ be the recorded relevance

grade, with an unjudged candidate assigned evaluation grade zero. Define $k_q = \min(k, P_q)$. Normalized discounted cumulative gain is

$$\mathrm{nDCG}_q\,@k = \frac{\sum_{i=1}^{k_q}(2^{r_q(d_i)}-1)/\log_2(i+1)}{\mathrm{IDCG}_q\,@k}. \tag{3}$$

For an ideal-reference set B_q, let $r^{\downarrow}_{q,B_q,i}$ denote its grades sorted in descending order. Ideal discounted cumulative gain (IDCG) is

$$\mathrm{IDCG}_q\,@k = \sum_{i=1}^{\min(k,|B_q|)} \frac{2^{r^{\downarrow}_{q,B_q,i}}-1}{\log_2(i+1)}. \tag{4}$$

The implementation returns zero when IDCG is nonpositive. Candidate-conditioned IDCG uses $B_q = D_q$; full-query IDCG uses the complete recorded query qrels, including judged documents outside the candidate pool. Thus a short candidate ranking does not truncate the full-query ideal to the pool size. Evaluation zeros for unjudged documents do not establish irrelevance.

For binary relevance $z_{qi} = \mathbf{1}\{r_q(d_i) > 0\}$ and $R_q = \sum_{i=1}^{P_q} z_{qi}$, the historical exact-versus-greedy endpoints are

$$\begin{aligned} \mathrm{AP}_q &= \frac{1}{R_q}\sum_{i=1}^{P_q} z_{qi}\frac{\sum_{j=1}^{i} z_{qj}}{i}, \\ \mathrm{RR}_q &= \frac{1}{\min\{i : z_{qi}=1\}}. \end{aligned} \tag{5}$$

Both are defined as zero when $R_q = 0$. Average precision (AP) and reciprocal rank (RR) use the complete recorded candidate ranking, not the full corpus. Within a single cell, mean average precision (MAP) and mean reciprocal rank (MRR) are the arithmetic means of AP and RR over eligible queries; the historical MAP label does not change AP's candidate-positive denominator. Within a fixed comparison, methods share candidates, labels, cutoff and IDCG. Across families, changing IDCG can rescale query effects and their aggregate weighting; the original contracts are retained.

We distinguish graph modification, full-ranking response and evaluated grade-position response. If a changed ranking has the same first-k grade sequence and fixed IDCG, Eq. (3) is unchanged. This is a sufficient explanation for a recorded zero; different grade sequences can also yield equal metric values. Top-k membership is a separate diagnostic: reordering a fixed set can change nDCG, while exchanging equal-gain documents can preserve it. We therefore do not use membership change as a necessary gate for retrieval activity.

4 Experimental Design

4.1 Evidence cohorts and reuse

Table 1 separates the evidence populations. We use relevance judgments (qrels) as the external evaluation reference. The canonical score-derived study uses SciDocs, FiQA, HotpotQA and BRIGHT [9, 23, 26, 34]. These benchmarks provide heterogeneous relevance supervision, including citation- and question-answering-derived labels. They are not uniformly direct human pairwise gold standards. Newer FiQA and SciDocs cohorts use BEIR-hosted snapshots [28]; the historical HotpotQA retrieval conversion remains incompletely pinned.

Canonical counts are 120 SciDocs, 120 FiQA, 52 HotpotQA and 50 BRIGHT queries. Eighteen of 70 stored HotpotQA queries fail the canonical relevant-candidate eligibility condition. The six-query primary cache is disjoint from the selected 213-query cache; the latter comprises 77 SciDocs, 76 FiQA, 42 BRIGHT and 18 HotpotQA identities. It is selected and heterogeneous rather than a random prospective replication. Seventy-nine selected-cache pools have no judged-positive candidate; they remain in principal means.

The three newer 20-query cohorts are mutually disjoint. BGE reuses the PRP and prospective pools; graph-construction variants and prompt development reuse the prospective queries. The prospective collection contains 5,880 responses, but only 20 query units. The offline replay contains 2,405 graph tasks and 56,335 metric rows; these workload counts create no additional independent queries. Online Resource 1 preserves cohort selection, snapshot and reuse details.

4.2 Construction, repair and extraction contracts

The canonical graph uses stored BM25 lexical-retrieval scores, term frequency–inverse document

Table 1 Evidence cohorts and comparison contracts. Counts are unique query identities within each row; rows are not an additive independent sample. C/F denote candidate-conditioned/full-query IDCG.

Cohort	Queries	Pool / IDCG	Selection and role
Score-derived	342	20 or 10; larger 50 or 35 / C, F	Stored multi-ranker scores; 1,025 repeated query–regime exact instances; C for repair–unrepaired, F for pooled exact–greedy; details in Section 4.3
Cached primary	6	8 / F	Qrel-enriched four-provider diagnostic; complete optimum envelope
Selected cache	213	15 / F	Retrospective provider-availability strata: 110 two-provider and 103 four-provider aggregates
PRP-Graph pilot	20	20 / F	10 FiQA, 10 SciDocs; adapted external acquisition/aggregation comparison
Prospective	20	12 / F	7 FiQA, 7 SciDocs, 6 BRIGHT; two compact models, strict repair contrasts
Candidate diagnostic	20	12 per arm / F	Further queries; BM25 versus BGE selection from BM25 top 100

frequency (TF-IDF) scores and MiniLM sentence-encoder scores as a controlled multi-source substrate. Per-query, per-ranker min–max normalization precedes margin votes. Retention-matched thresholds depend on stored score distributions, not relevance labels, but are calibrated on the evaluated query list and are therefore transductive. The permissive `ms1` rule retains ranker-supported directions; `ms2` requires support from at least two of three rankers; `ms1_drop_mutual` removes contested pairs during construction. This last rule differs from deleting reciprocal edges from an already constructed graph. Candidates come from the fixed reciprocal-rank-fusion pool policy; full thresholds and alternative policies are in Online Resource 1.

For cached judgments, strict parsing accepts complete A/B choices, maps reversed presentations back to document identity, and retains a provider edge only when both valid orientations agree. AB denotes the original document presentation and BA its reversal. Invalidity, abstention and reversal conflict never become a default winner. Individual-provider edges have unit weight; the primary aggregate sums contributions of one quarter per provider, preserving opposing directions. All candidates, including isolates, remain nodes. Prospective Qwen2.5-7B-Instruct (7 billion parameters) and SmolLM3-3B (3 billion parameters) judgments use a separately frozen JSON protocol; strict construction similarly requires agreement of valid directional responses. Confidence labels are not calibrated probabilities. Alternative constructions are exploratory repeated views, not selected replacements for the strict protocol.

Greedy repair repeatedly deletes a minimum-weight edge from a deterministic depth-first-search cycle until acyclic. It is not the Eades–Lin–Smyth ordering heuristic [13]. Exact repair uses a linear-ordering mixed-integer program solved by SCIP through PySCIPOpt [1, 22], requiring proven optimality with zero reported gap. Pool, evidence, extractor, ties and cutoff remain fixed within a contrast, as shown in Fig. 1. The complete six-query diagnostic enumerates all distinct optimal deletion sets with independent completeness checks; it is conditional on six eight-node graphs, not a new optimizer.

External comparisons include original-graph SpringRank, direct Bradley–Terry-style aggregation, graph-free fusion, adapted PRP-Graph and a trained BGE cross-encoder. BGE follows the official raw-relevance-logit procedure on matched documents [4]; it is not budget- or training-matched to a generative comparator. The adapted PRP pilot retains the published implementation's Swiss comparisons and personalized PageRank,

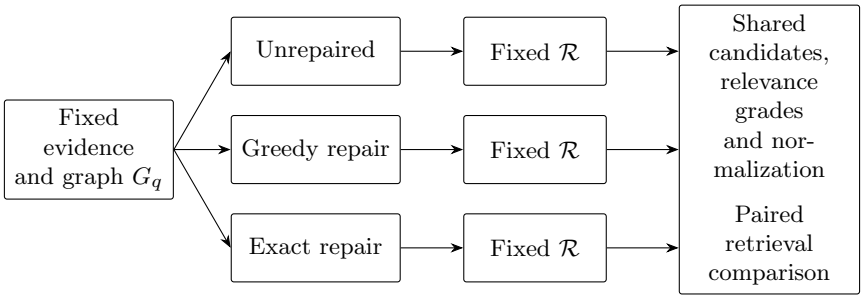


Fig. 1 Matched repair comparison. Each branch uses the same deterministic extractor $\mathcal{R}$, candidate set, tie handling, relevance grades, evaluation cutoff and normalization. Only graph repair changes; alternative aggregation methods are separate controls

but changes local pooling and runtime orchestration. It is not an exact reproduction of the published retrieval experiment. Missing or invalid listwise outputs remain missing controls. These distinctions prevent an acquisition or extractor change from being attributed to edge deletion.

4.3 Evaluation and query-level inference

The greedy-versus-unrepaired 20/60/110 and exact-versus-unrepaired 36/56 families use candidate-conditioned nDCG. Canonical cutoffs equal pool size: 20 for SciDocs, FiQA and BRIGHT, and 10 for HotpotQA. The larger-pool study retains its recorded (P, k) cells, including $P = k$ reference cells. The historical exact-versus-greedy 35/399 families instead use full-query IDCG at eligible cutoffs 5, 10 and 20; their MAP and MRR use complete candidate rankings. Cached/prospective LLM, SpringRank, PRP-Graph, BGE and candidate-coverage results use full-query nDCG@10. These families' absolute means are not directly interchangeable. Online Resource 1 provides the endpoint/family map.

The corrected 35-test family crosses seven fixed extractors—balance, balance hybrid, Copeland, Copeland hybrid, damped Markov, topological and priority-topological—with MAP, MRR and the three nDCG cutoffs. The independent unit is the dataset–query identity, conditional on the benchmark sampling assumptions. Within a repair-versus-unrepaired cell, inference uses one paired query effect. In the separate pooled exact-versus-greedy diagnostic, repeated construction regimes form one dependent query block. Ordinary single-cell means weight eligible queries equally. The pooled diagnostic retains its original regime-row-weighted effect rather than silently substituting an equal-query average. For query sums S_q and eligible regime counts m_q, its statistic is

$$T = \frac{\sum_q S_q}{\sum_q m_q}, \qquad S_q = \sum_{r \text{ eligible}} \left(U_{qr}^{\text{exact}} - U_{qr}^{\text{greedy}}\right). \tag{6}$$

There are 342 query units for 28 comparisons and 290 for seven nDCG@20 comparisons, which exclude the original ineligible HotpotQA cutoff. Their 1,025 and 869 eligible regime rows, respectively, are not independent observations. The historically omitted empty BRIGHT/`ms2` record remains excluded, without zero imputation.

The completed correction uses 10,000 Monte Carlo whole-query sign assignments, seed 17, add-one two-sided p-values, and Holm correction over the unchanged 35-test family [15]. Bootstrap sensitivity uses 10,000 within-dataset query-cluster draws, seed 13, with percentile and basic 95% intervals. This corrected family does not use exact enumeration or bias-corrected and accelerated (BCa) intervals. All original effect estimates are preserved; historical row-based p-values are superseded.

Sign-flip inference assumes independent query units and exchangeability of the entire query's exact/greedy outcome vector under label swap. Mean zero alone does not justify that null, and the deterministic methods were not randomly assigned. Bootstrap intervals are approximate and pointwise, especially with sparse, skewed differences. Separate canonical, larger-pool, 399-cell

and exploratory families retain their own procedures and error scope; their juxtaposition provides no global error guarantee. In model-averaged prospective contrasts, fixed-model effects are averaged within query before inference; model-specific comparisons retain one effect per query. Non-significance is not equivalence, and a pointwise interval does not override a Holm decision. Online Resource 1 provides the complete saved families and correction details.

4.4 Evidence access and AI assistance

The supporting *Consistency-Aware Judgments v1* resource was publicly released before this revision. Its 56,332 records comprise 48,777 pairwise judgments, 7,140 repeated-measurement summaries, 139 selector-training records and 276 provider diagnostics. These are four configurations of one heterogeneous resource, not four independent datasets or 56,332 independent queries. The release excludes FiQA, Cohere and raw benchmark text; it is not the complete source of all cached or prospective results reported here. October prospective responses are not part of that release.

The unchanged confidential reviewer ZIP (35 files) independently reproduces all 35 corrected comparisons from processed paired observations (Level 2), including 342/290 query populations and zero Holm-adjusted rejections. Direct checks of 1,025 weighted graphs confirm 646 acyclic and 379 cyclic inputs and saved greedy deletion feasibility/cost. Other families retain saved-p-value checks; saved exact statuses are not new optimality proofs. A separate recovery audit locates the complete frozen selected-cache permutations and relevance maps and independently reconstructs all 426 branch attributions on 213 queries, including nDCG arithmetic. Complete frozen processed inputs also permit an independent, solver-free exhaustive reconstruction of the six-graph optimum envelope. These checks confirm existing results, not new observations or regenerated provider judgments. Historical score files and canonical qrels match recorded hashes, but the exact original MiniLM weight revision and historical exact-ranking permutations remain incompletely preserved. Online Resource 1 distinguishes these verification levels.

The recovered provider-linked graphs, permutations and relevance projections required to verify the central attribution and envelope analyses remain outside the 35-file reviewer ZIP pending applicable sharing clearance. Data-free offline verification procedures accompany the JDSA preparation sources; their input projections are not public artifacts. No raw provider transcripts, benchmark text, qrels or complete envelope inputs are added to that ZIP. Its content-specific audit covers proposed confidential review use, not full repository history or excluded material. Confidential delivery remains pending confirmation of a non-public editorial channel. The project repository is private; its address is https://github.com/SoroushVahidi/consistency-aware-llm-rankin. Public access is not claimed. Previously released resources and their coverage limits are identified in Online Resource 1.

Generative-AI assistance.

Anthropic Claude and OpenAI ChatGPT/Codex were used for software-development assistance, analysis and audit utilities, figure preparation, and drafting and revising portions of the manuscript. Scientific responsibility for the final text, code, analyses, figures and citations includes human verification against source materials and underlying evidence. No generative-AI tool is credited as an author.

5 Structural and Ranking Opportunities

5.1 Construction-dependent cyclicity

Table 2 distinguishes graph workload from query support. Of 1,025 canonical graph records, 646 are acyclic and 379 cyclic. The records reuse 342 queries across three regimes; 316 identities are cyclic under at least one regime. Conservative `ms2` graphs are acyclic throughout the observed cohort. This is an empirical property of these constructed inputs, not a theorem that majority-supported preferences are always acyclic.

In the active `ms1` regime, cyclic-query rates are 99.2% for SciDocs, 98.3% for FiQA, 63.5% for HotpotQA and 92.0% for BRIGHT. Construction-time mutual dropping reduces them to 11.7%,

30.8%, 1.9% and 22.0%. Post-hoc reciprocal deletion from the SciDocs `ms1` graph instead gives 10.8%; the different operation and threshold contract matter. Source-scale sensitivity is also substantial: conditional BM25 edge-weight share is 0.988 under raw margins and 0.512 after normalization. These measurements establish construction dependence, not the semantic accuracy of the resulting votes.

The genuine-judgment cohorts demonstrate a separate aggregation effect. All 24 individual-provider graphs in the six-query corrected pilot are acyclic, but all six four-provider aggregate graphs are cyclic. In the selected cache, the two disjoint provider-availability strata contain 89/110 and 102/103 cyclic aggregate graphs. Absence of structural opportunity therefore cannot explain the selected-cache outcomes generally.

The prospective strict protocol has no cyclic-positive case: only one of 40 model/query graphs is cyclic, and its candidate pool contains no judged-positive document. All 14 positive-containing model/query graphs are acyclic. The alternative constructions create more cycles but reuse the same 20 queries. Qwen cyclic counts for strict, AB-only, BA-only, symmetric-both-valid and symmetric-available rules are 0/8/8/5/5; Smol counts are 1/13/12/1/5. More retained edges change the intervention opportunity, without establishing more reliable preference evidence.

5.2 Repair activity and ranking response

Both repairs modify all 379 cyclic canonical records and all 191 selected-cache cyclic aggregates. Exact repair has a different deletion set from greedy in 87.9% of canonical cyclic records. It strictly lowers removed weight in 333 of those 379 records; lower dataset-mean removal costs are shown in Fig. 2. Mean reductions relative to greedy are 25.1%, 29.2%, 13.4% and 22.3% for SciDocs, FiQA, HotpotQA and BRIGHT. These are structural quantities; retrieval effect sizes are evaluated separately.

Graph activity does not imply a full-ranking response. Among the 191 cyclic selected-cache graphs, greedy leaves 11 Copeland rankings unchanged and exact leaves 31 unchanged. Including 22 acyclic graphs, unchanged-ranking totals are 33 and 53 of 213. This difference shows that choosing a structurally cheaper deletion can change the amount of extracted-order activity; it is not evidence that activity itself is beneficial. In the six-query aggregate diagnostic, each repair changes two rankings and leaves four unchanged.

The prospective construction workload contains 400 greedy/exact branch records across 200 model/query/construction graphs. Existing diagnostics identify 116 graph changes, 106 ranking changes and nine nonzero utility records, covering six query identities. These workload counts are descriptive repeated measurements of 20 queries. Complete historical permutations are unavailable for the 1,025 historical exact canonical scalar records, so a corresponding full canonical rank-response decomposition is not reconstructed. The selected-cache and prospective decompositions are therefore reported with their own explicit support rather than extrapolated to all canonical graphs.

6 Relevance-Profile Attribution and Its Composition

Table 3 attributes the selected cache's ranking-change zeros using stored Copeland permutations and the frozen relevance map. No new historical model/provider generation or ranking acquisition is performed; discounted-gain and nDCG arithmetic are independently recomputed from the preserved records. All 213 identities enter both repair branches. For exact repair, 160 rankings change: 135 retain recorded nDCG, comprising 61 zero-positive pools and 74 positive-containing pools with identical evaluated grade sequences. Greedy changes 180 rankings; its 143 zero-effect cases split into 69 and 74. No positive-pool changed-ranking zero remains with a different top-ten grade sequence in this selected cohort.

Within the 74 positive-pool equal-profile cases, all moved documents are unjudged in 55 greedy and 58 exact records. Positive documents move in six and five records, respectively, without changing the evaluated grade sequence. Movement outside the prefix or between equal-grade positions can preserve that sequence; those subtypes were not separately classified. The attribution explains the realized metric under its qrel contract, not the semantic reason for the comparisons or the best utility obtainable from a different intervention.

Table 2 Structural and ranking opportunity. Greedy/exact outcomes share queries. A dash denotes an unavailable complete ranking decomposition, not zero activity

Cohort	Query IDs	Graph records	Cyclic	Cyclic, same rank G/E
Canonical	342	1,025	379	—
Selected cache	213	213	191	11 / 31
Strict prospective	20	40	1	0 / 0

Canonical records repeat construction regimes. Prospective records repeat two models; its sole cyclic graph has no judged-positive candidate. Both repairs change every reported cyclic graph in these rows.

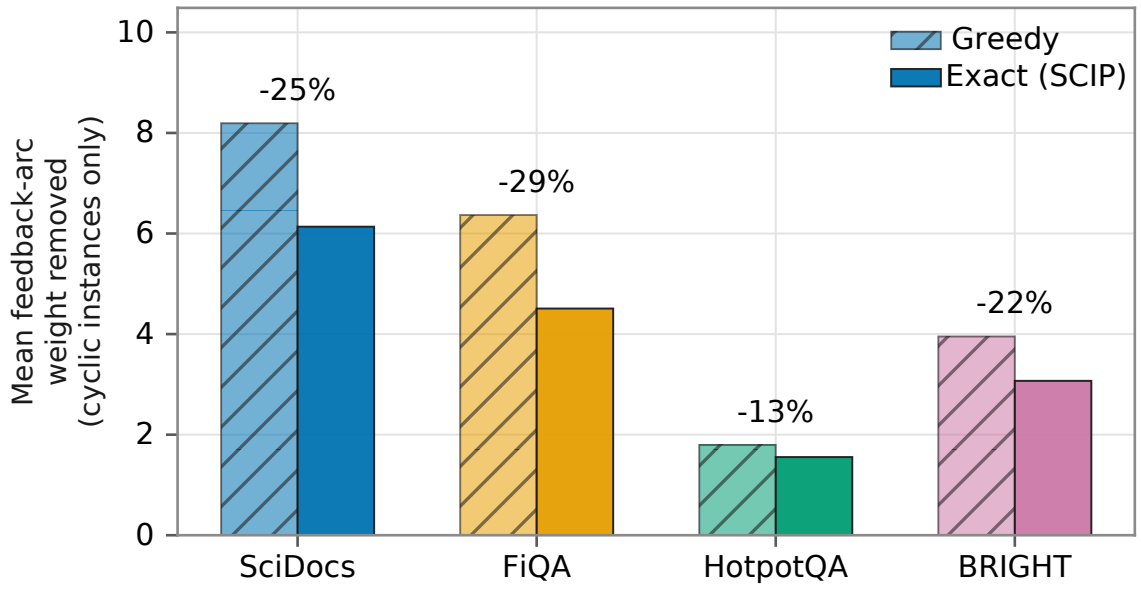


Fig. 2 Mean deletion weight on 379 cyclic canonical query–regime records, pooled over ms1, ms2 and construction-time mutual dropping. Hatched/solid bars denote greedy/exact repair. Percentage labels show the original rounded reduction from the greedy dataset mean; individual records can tie. These are structural costs, not retrieval effects

Top-ten membership changes in 56 greedy and 38 exact cases, yet 45 and 32 still have zero measured effect. Conversely, 26 greedy and 19 exact nonzero effects occur without membership changes. In larger canonical pools, the mean membership-change rate is 10.6% over dependent query–regime–cell rows. Membership activity therefore supplies neither a necessary nor a sufficient indicator of relevance gain.

The four symmetric-available cyclic-positive prospective cells illustrate the same measured explanation on only three query identities. Both repairs change full rankings, but every moved document is unjudged and the top-ten grade sequence remains fixed. These selected solutions are descriptive cases, not evidence that all equally optimal repairs or all reliable comparison settings are utility-inert. Directional constructions provide contrary cases: nine nonzero records include harms and one exact improvement, six within an unchanged top-ten set. All eight model-averaged exploratory construction contrasts retain Holm p=1.

Table 3 Saved relevance-profile attribution on 213 selected query identities. Mutually exclusive rows sum to 213 in each branch. The two columns share queries; the 74-case subsets are not 148 independent observations

Saved outcome	Greedy	Exact
Full ranking unchanged	33	53
Ranking changed; no judged positive in pool	69	61
Ranking changed; positive pool; same evaluated top-ten grade sequence and zero metric change	74	74
Ranking and recorded metric changed	37	25

Metric-active cases contain 15 gains/22 harms for greedy and 16 gains/9 harms for exact. Unchanged rankings include 22 acyclic graphs and 11/31 cyclic graph-changing cases.

6.1 Composition of observed opportunity losses

Table 4 keeps the distinct-query denominator visible and separates dataset from provider-availability descriptions. Of the 79 zero-positive pools, 36 are SciDocs and 38 FiQA: these two datasets contribute 74/79, compared with four BRIGHT and one HotpotQA pool. Absence of judged positives is therefore composition-dependent. By contrast, positive-pool fixed-profile zeros occur in all four datasets and both availability strata. They cannot be attributed entirely to a single dataset or to missing positives. These are observed evaluation mechanisms, not explanations of semantic relevance or preference correctness.

Of 134 positive-containing pools, 120 are cyclic. Greedy changes 111 positive-pool rankings, with 37 measured changes; exact changes 99, with 25 measured changes. The remaining 74 per branch retain the evaluated profile. Among 79 zero-positive pools, greedy/exact change 69/61 rankings without changing measured utility; 10/18 rankings remain unchanged. Thus the all-query and changed-ranking denominators answer different questions. The corresponding all-query counts, including unfavorable cases, remain the principal description; conditioning does not create new evidence of efficacy.

The two-provider stratum has 57 cyclic-positive queries among 110, versus 63 among 103 in the four-provider stratum. Dataset, available providers and retained evidence vary together; these strata are not randomized treatments or isolated provider interventions. The complete dataset × availability × positive-availability table in Online Resource 1 includes an empty HotpotQA/two-provider/zero-positive cell and zero-positive cells of size one or two. These cells support counts only, without stable subgroup-effect claims. No stratum was selected for significance and no testing family is added.

Observed fixed-profile case (C1).

In an authentic BRIGHT 2-provider selected-cache record, greedy repair exchanges the documents at ranks 2 and 5. All moved documents are unjudged. The evaluated top-ten grade vector remains $(1, 0, 0, 1, 0, 0, 0, 0, 0, 0)$, and full-query nDCG@10 remains 0.558508 (change zero at 10^{-12} tolerance). This is an observed saved-ranking case, not a synthetic permutation; its source linkage is preserved privately. Unjudged is an evaluation status, not proven irrelevance. Section 8 gives the contrasting observed case in which an optimal-repair choice changes a positive document's rank and measured utility.

7 Retrieval Effects and Uncertainty

7.1 Retrieval effects under matched repair

Table 5 summarizes the principal matched families. The canonical active and full greedy families have zero Holm rejections, as does the 110-cell larger-pool family. Online Resource 1 retains the original active-regime forest display with pointwise uncertainty; Table 6 brings saved effect estimates and informative query counts

Table 4 Composition of the 213 selected queries. $P+$ counts pools with a judged-positive candidate. Paired entries are greedy/exact (G/E) counts on the same queries. Same/changed refer to full rankings; Eq.+ counts changed-ranking, positive-pool cases with the same evaluated top-ten grade profile and zero metric change. ΔU counts measured utility changes. Dataset and availability panels are overlapping descriptions, not independent samples.

Group	N	$P+$	Same G/E	Changed G/E	Eq.+ G/E	ΔU G/E	Gains G/E	Harms G/E
SciDocs	77	41	10/15	67/62	16/23	19/9	7/7	12/2
FiQA	76	38	6/18	70/58	27/22	8/7	5/4	3/3
BRIGHT	42	38	14/16	28/26	18/18	8/6	2/3	6/3
HotpotQA	18	17	3/4	15/14	13/11	2/3	1/2	1/1
Two-provider	110	71	30/49	80/61	31/26	19/12	8/7	11/5
Four-provider	103	63	3/4	100/99	43/48	18/13	7/9	11/4
All queries	213	134	33/53	180/160	74/74	37/25	15/16	22/9

into the main text. Exact-versus-unrepaired families likewise retain zero rejections. These are no-demonstrated-benefit findings under the stated protocols and sensitivity, not evidence that all true effects vanish.

7.2 Informative support and pointwise uncertainty

The 35 corrected exact-versus-greedy comparisons contain 28 endpoints with 342 independent query units and seven with 290; only 4–99 query sums per endpoint are nonzero. Table 6 shows saved examples alongside both selected-cache strata and the six-query primary diagnostic. Its intervals are pointwise sensitivity summaries under their original family-specific procedures, not simultaneous confidence intervals. No pooled inferential family or altered weighting is introduced. Copeland's percentile interval excludes zero while its basic interval includes zero; the respective Holm decision remains no rejection. Improving deletion cost does not demonstrate a retrieval advantage in the corrected 35-test diagnostic. It has zero Holm rejections; the minimum adjusted p is 0.85041496. Priority-topological nDCG@10 has mean exact-minus-greedy effect −0.002229580 across 342 query units, raw p=0.02429757 and Holm p=0.85041496. Its query sums are positive/negative/zero for 30/44/268 identities. A pointwise percentile interval [−0.004356624,−0.000334221] does not establish a family-corrected discovery. The separate saved 399-cell per-dataset/regime family also reports zero rejections and was not rerun in the correction.

The corrected six-query primary aggregate has mean nDCG 0.881750 unrepaired and 0.882760 after either selected repair, a difference of +0.001010 with saved interval [0,0.003030]. One query improves and five tie; raw and adjusted p-values equal one. This ten-test family cannot attain its first Holm threshold with only six queries even if every query changes: the smallest attainable two-sided exact sign-flip p-value is $2/2^6 = 0.03125$, exceeding $0.05/10 = 0.005$. In the selected cache, exact means improve by +0.003164 and +0.004840 in the 110/103 strata, each with Holm p=1 in the existing exploratory family. Greedy differences are −0.000950 and −0.002248. Positive selected point estimates do not establish a general benefit.

Complementary external controls help assess whether the repaired pipeline earns its complexity. On 213 selected queries, SpringRank, unrepaired Copeland, greedy Copeland and exact Copeland means are 0.357436, 0.353838, 0.352261 and 0.357813; exact exceeds SpringRank by 0.000377. None of the eight matched SpringRank contrasts survives Holm correction. Table 7 adds the trained BGE and adapted PRP-Graph controls. Training, model family, acquisition and compute differ; the table is matched-document effectiveness context, not an isolated deletion test or a confirmatory ranking of model families.

BGE's exploratory paired differences on the PRP pool are +0.075682 versus BM25, interval [−0.004973,0.158395], and +0.084332 versus

Table 5 Principal matched result families. Test counts are hypotheses, not query counts. Separate rows do not define a combined family-wise guarantee

Contrast / scope	Query support	Tests	Holm rejections
Greedy–unrepaired, active canonical	50–120 per dataset	20	0
Greedy–unrepaired, full canonical	342 roster, repeated cells	60	0
Greedy–unrepaired, larger pools	342 roster, repeated cells	110	0
Exact–unrepaired, canonical	342 roster	36	0
Exact–unrepaired, larger pools	342 roster	56	0
Exact–greedy, corrected pooled	342; 290 at nDCG@20	35	0

Table 6 Saved nDCG@10 effects and pointwise 95% intervals. N is distinct queries; $n_{\neq 0}$ counts nonzero query sums for the pooled corrected family and nonzero query differences otherwise. The pooled exact-minus-greedy estimand retains regime-row weighting. Other rows use their original query means. Intervals are percentile except where basic is explicitly shown; Explor. denotes the saved exploratory family; each Holm decision belongs only to the stated original family. These are selected informative displays, not a new testing family.

Comparison	Family	N	$n_{\neq 0}$	Mean Δ	Pointwise interval	Holm p
Copeland E–G	Corrected 35	342	63	−0.001785	[−0.003584, −0.000112]	1
basic sensitivity	same endpoint	342	63	same	[−0.003459, 0.000013]	1
Priority-topological E–G	Corrected 35	342	74	−0.002230	[−0.004357, −0.000334]	0.850415
G–U, two-provider	Explor. 502	110	19	−0.000950	[−0.007574, 0.005823]	1
E–U, two-provider	Explor. 502	110	12	0.003164	[0.000060, 0.007053]	1
G–U, four-provider	Explor. 502	103	18	−0.002248	[−0.014272, 0.009574]	1
E–U, four-provider	Explor. 502	103	13	0.004840	[−0.001427, 0.014088]	1
G/E–U, primary aggregate	Primary 10	6	1	0.001010	[0, 0.003030]	1

adapted PRP-Graph, interval [0.008773,0.180777]. These pointwise, unregistered comparisons are not confirmatory superiority tests. The adapted PRP comparison's eight primary native/random-acquisition contrasts all have Holm p=1. The strict prospective repair effects are zero for every query; the inactive opportunity documented above limits that null's interpretation. Invalid direct-listwise outputs, zero valid Qwen and ten valid Smol rankings among 20 attempts each, cannot serve as complete baselines or be scored zero.

Table 7 Completed external controls: full-query nDCG@10 on two disjoint 20-query pools. Shared BM25/BGE prospective values are not replicated observations. U/G/E denote unrepaired/greedy/exact Copeland

Method	PRP pool	Prospective Qwen	Prospective Smol
BM25	0.278865	0.080112	0.080112
BGE cross-encoder	0.354547	0.108610	0.108610
Adapted PRP-Graph	0.270215	—	—
U-Copeland	0.271771	0.126872	0.097435
G-Copeland	0.271771	0.126872	0.097435
E-Copeland	0.271526	0.126872	0.097435
SpringRank	0.259999	0.134091	0.085859
Bradley–Terry-style	0.254006	0.134091	0.089653

7.3 Limits of the completed prospective instrument

The strict prospective null must be read with the instrument's observed dispositions. Qwen abstains on 1,378 of 2,920 valid pairwise outputs. Smol selects displayed A on 2,591 of 2,772 directional responses, despite reversed presentation. There are 148 capped invalid outputs; these belong to invalid-response accounting, not an additional independent sample. Abstentions, invalids and reversal conflicts do not become directional votes. Repeated observations assess reliability without adding primary votes or query units. Parsing success alone therefore does not establish an informative instrument.

Strict construction produces no cyclic-positive graph: its sole cyclic model/query graph has no judged-positive candidate, and all 14 positive-containing model/query graphs are acyclic. The resulting zero repair effects test an almost inactive intervention under a compromised instrument, not equivalence in reliable cyclic-positive evidence. Invalid listwise attempts remain missing controls (20/20 Qwen and 10/20 Smol), without zero imputation. Full dispositions, model-specific repeats and the unsuccessful neutral-schema development are retained in Online Resource 1. No results from the separately scheduled replication enter this article.

8 Finite Optimal-Repair Diagnostic

The six-query complete envelope supplies a different diagnostic: structural cost does not identify external utility even among optimal deletion sets. Table 8 retains all six instances. Four have a unique optimum. FiQA/2377 has two optimal repairs exchanging grade-one documents at ranks two and three, with unchanged utility. One SciDocs graph has two cost-0.75 repairs that exchange a grade-one and a grade-zero document: placing the positive at rank seven gives 0.956591277, while rank eight gives 0.950531008.

This finite range over a complete optimum set is not a confidence interval. The certified mean utility range is [0.881749864,0.882759909], corresponding to improvement [0,+0.001010045] over unrepaired Copeland. The selected SCIP optimum is already utility-best on the affected query. An equally optimal choice removes the pilot's sole improvement and leaves six ties; no certified choice produces a negative effect or strict sign reversal. The 0.006060269 per-query width demonstrates conditional non-identification, not a population effect, new algorithm or deployable relevance-aware selector.

9 Discussion and Limitations

9.1 What the audit establishes

The evidence distinguishes intervention inactivity from outcome inactivity. Construction eliminates the need for deletion in many canonical records and almost the entire strict prospective

Table 8 Certified utility ranges over all optimal repairs of six primary graphs. SciDocs identifiers are shortened; complete identities, certificates and full-precision utilities remain in the study archives, outside the reviewer ZIP. The reported selected SCIP repairs attain the maxima; utilities are rounded to six decimals here

Dataset / query	Cost	Repairs	Min nDCG	Max nDCG
FiQA /11039	1.50	1	0.835891	0.835891
FiQA /2377	1.00	2	0.727330	0.727330
FiQA /4002	0.25	1	0.776747	0.776747
SciDocs /067405...	0.25	1	1.000000	1.000000
SciDocs /1b1aed...	0.50	1	1.000000	1.000000
SciDocs /970093...	0.75	2	0.950531	0.956591

collection. That explanation is insufficient for the selected cache, where most aggregates are cyclic. Some deletions leave the fixed extractor's order unchanged; many changed orders leave the evaluated gain positions unchanged. The relevance-profile attribution explains these measured zeros more precisely than a statement that repair merely lacks statistical significance.

These findings also distinguish two kinds of exactness. A solver can certify minimum disagreement cost without identifying retrieval utility. Complete enumeration can identify the utility range conditional on one small frozen graph, but supplies no population guarantee. The latter diagnostic shows why a single returned optimum is not a sensitivity envelope. Neither result establishes a universal monotone relationship, or a universal absence of association, between structural improvements and retrieval effects.

9.2 Implications for evaluation practice

For data science and analytics, the practical implication is to validate the path from an optimized graph property to the downstream quantity of interest. Repair evaluations should report the candidate/relevance contract, judgment dispositions, cyclicity, graph modification, extracted-order response and evaluated utility separately. A matched unrepaired control is essential; direct graph rankers and trained relevance baselines answer complementary pipeline questions. Query identities, rather than responses or graph variants, determine inferential support. Where exact solving is feasible, it can control deletion-cost suboptimality; its output should not be described as relevance-optimal.

These are methodological checks, not a validated rule for choosing repair on an unseen query. Historical selector and mixed-protocol headroom analyses in this research program have not established a generalizing utility selector. They also do not prove one impossible. Positive and harmful cases must remain visible when an aggregate result is small. A benchmark zero caused by missing positives or fixed evaluated grades should be interpreted as a limitation of realized measured opportunity, not automatically as evidence that the underlying preference information has no value.

9.3 Measurement, sampling and inferential limitations

The main canonical evidence uses shallow score-derived graphs, while the selected cache is heterogeneous and nonrandom. The six-query primary pools are qrel-enriched. The prospective compact-model collection avoids that selection mechanism but has no strict cyclic-positive case; it is weak evidence about useful repair with reliable contradictory preferences. Alternative graph constructions reuse the same queries and change evidential quality as well as cyclicity. Format validity and repeatability do not establish semantic correctness, and the completed neutral-schema development did not validate a better instrument.

Relevance coverage is incomplete. Unjudged movements dominate many equal-profile cases, and zero evaluation gain does not prove true irrelevance. A separate 20-query candidate diagnostic

increases judged-positive coverage from 6 to 8 pools and the oracle ceiling by +0.069106, with interval [−0.016958,0.164250], but loses coverage in one domain. This is uncertain measured headroom, not an achieved ranking benefit. Neither that diagnostic nor the profile attribution rules out better candidates, alternative ranking policies or useful preferences in unjudged documents.

Inferential conclusions remain assumption-dependent and limited by sparse changes, modest query support and distinct families. The saved canonical finite-cell median minimum detectable effect is 0.0201 versus observed absolute mean effect 0.0036; sensitivity is not a bound on the true effect or a practical-importance threshold. Only a minority of saved equivalence cells meet their declared margins. We do not infer blanket equivalence or recover missing evidence by changing the independent unit, tuning outcomes or combining incompatible normalization contracts.

Finally, complete historical exact ranking permutations are unavailable for the canonical scalar records, and the full optimum envelope covers only six eight-node graphs. Local reconstruction of attribution from saved permutations and of this finite envelope strengthens verification within those cohorts; it does not reproduce historical model serving, recover missing permutations or broaden the sampling scope. Existing external baselines differ in model training, truncation and compute; they are contextual controls, not a comprehensive ranking of modern retrieval systems. Historical provider aliases and some dataset conversions are incompletely pinned; restrictions limit excluded historical inputs. The final reviewer ZIP supports independent processed-observation statistics and graph checks, with historical reconstruction limits and editorial-channel confirmation still required.

10 Conclusion

Preference-graph repair can succeed structurally while failing to change an extracted order or its evaluated relevance-grade positions. The observed opportunity losses differ across protocols: many canonical graphs and almost all strict prospective graphs are acyclic, whereas the selected cache contains substantial structural and ranking activity with frequent measured utility inactivity. Relevance-profile attribution explains its changed-ranking zeros under the frozen relevance contract. None of the evaluated 35 exact-versus-greedy comparisons rejects after its specified Holm correction, and a complete small-instance envelope shows residual utility ambiguity among equal-cost optima. These findings justify separate reporting of structural, ranking and retrieval outcomes. Selected cohorts, incomplete judgments, limited informative opportunities and inferential assumptions prevent a universal negative conclusion, statistical equivalence or a validated repair-selection rule.

Supplementary information

Online Resource 1 (ESM_1.pdf): protocols, query populations, complete saved comparison families, structural and retrieval diagnostics, source identities, reuse and reproducibility limits.

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Statements and Declarations

Data availability.

Public benchmark sources are SciDocs, FiQA, HotpotQA and BRIGHT, with the exact source and conversion limits given in Online Resource 1. The previously released derived judgment and ranking-output resources are linked there with their exclusions. They do not supply all current experimental inputs. The existing confidential reviewer package contains processed paired observations and weighted graphs; prior independent Level 2 validation reproduces all 35 corrected comparisons, including 342/290 query support and zero Holm rejections. Original scoring, ranking/grade attribution and the six-graph optimum envelope are not fully reconstructible from that package. Complete frozen private inputs separately support independent reconstruction of the selected-cache attribution and six-graph finite envelope; their recovered numerical projections remain excluded pending applicable sharing clearance. Historical score/qrel hashes

are verified, without claiming exact original scorer/model regeneration. Raw benchmark text, source responses, raw qrels, historical permutations and complete envelope inputs are excluded. The verified package is designated for confidential review only; delivery through a confirmed non-public editorial attachment category remains pending.

Code availability.

The project code is maintained at https://github.com/SoroushVahidi/consistency-aware-llm-rankin, currently private. Public access is not claimed. The bounded processed-evidence verification scripts remain in the confidential reviewer package (35 files; checksum in Online Resource 1). Installation instructions and source provenance accompany the JDSA preparation package; full reproduction remains limited by the unavailable or restricted inputs above.

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Competing interests.

The author declares no competing interests relevant to this work.

Author contributions.

Soroush Vahidi conceived the study, designed and implemented the experimental pipeline, conducted the original experiments, analysed the results and prepared the manuscript and figures. The present revision uses existing evidence. Anthropic Claude and OpenAI ChatGPT/Codex assisted with software development, auditing, figure preparation and drafting/revision, as disclosed in Experimental Design. The author confirmed these contributions and accountability on 26 October 2026 and retains responsibility for final content and approval; the automated editorial checks are not represented as independent human verification of every assisted output.

Ethics and consent.

No new human participants or human relevance labels were collected. The study uses existing benchmarks and model outputs. Participant consent and consent for publication of individual personal data are not applicable.

AI-assisted tools.

Anthropic Claude and OpenAI ChatGPT/Codex assisted with development, audit utilities, figures and drafting/revision, including this journal conversion, as described in Section 4.4. The author retains responsibility for verification and final approval. No new model judgments or experiments were generated for this revision.

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